COIL PROJECT

Sentimental Analysis chatbot

TEAM MEMBERS

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OVERVIEW

Introduction

Data pre-processing

Data analysis

Methodology

Evaluation

INTRODUCTION

1

Sentimental analysis

- A chatbot using NLP
- Determine the emotions
- Chance to enhance bussiness

2

Data overview

- 417000 different data with no missing or unmatched data
- Nominal data with 6 categories

3

DATA PREPROCESSING

Eliminating extra spaces

Expanding contractions

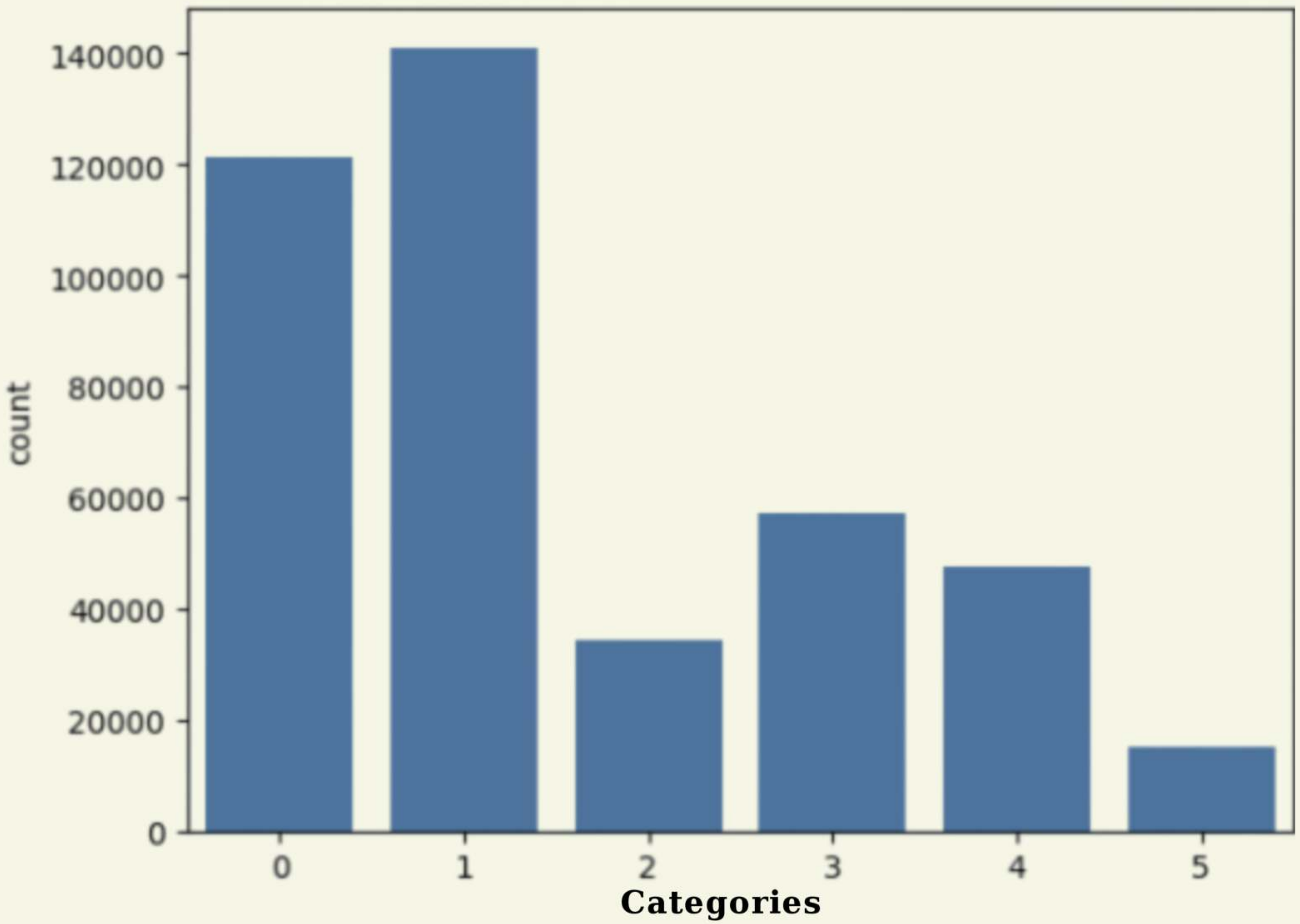
Removing usernames

Removing stop words

Lemmatizing

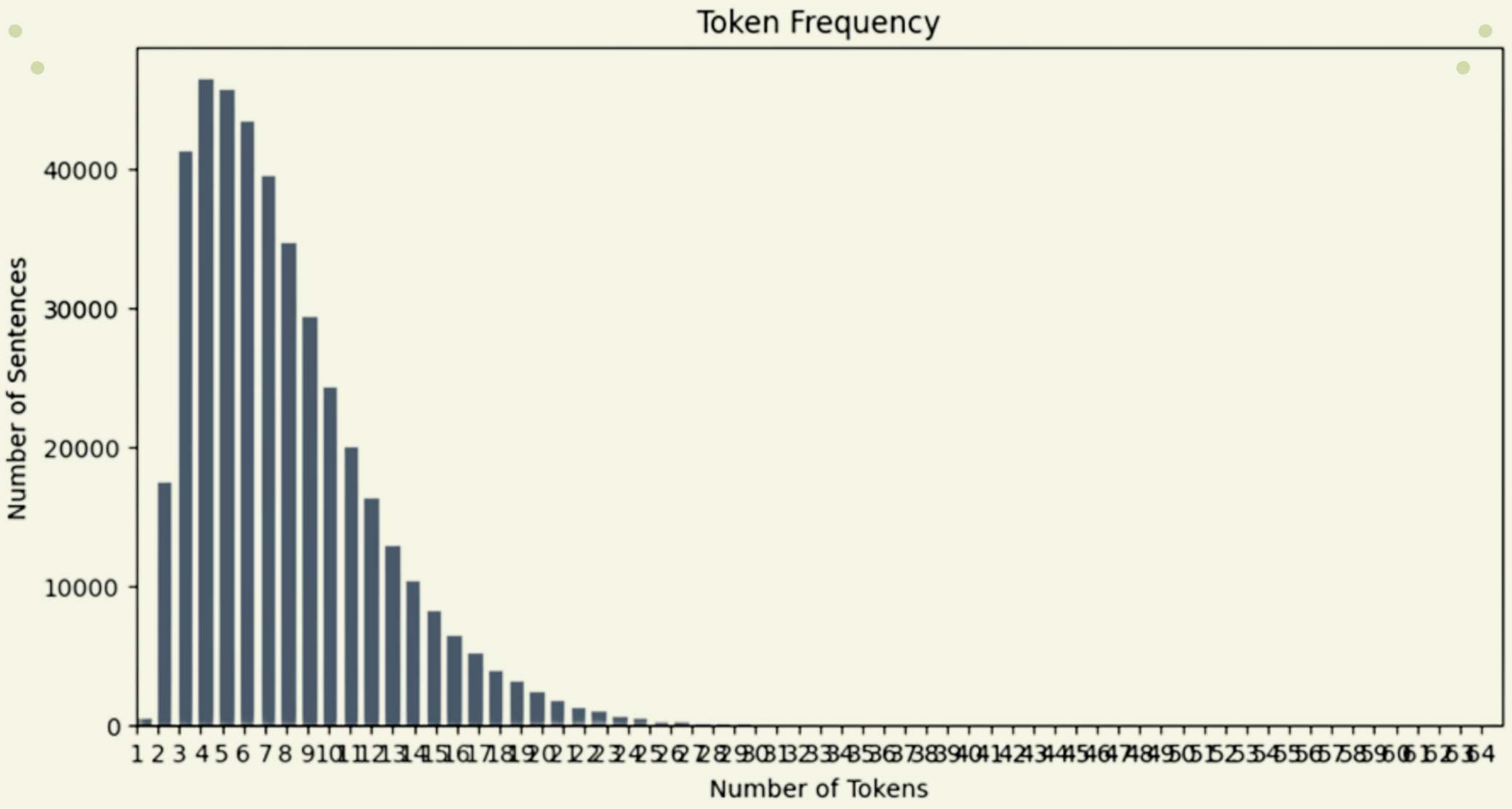
Checking and fixing miss spelled words

DATA ANALYSIS



Bar chart of label counting

DATA ANALYSIS



Histogram of token length

DATA ANALYSIS

Labels	Categories	Unique worlds
0	Sadness	abuse, ache, awful, awkward, boring, burden, completely, defeat, depressed, disappointed, discourage, drain, dull, embarrass, emotional, exhaust, guilty
1	Joy	accepted, bless, calm, comfortable, confident, content, cool, creative, cute, enjoy, excite, free, glad, important, inspire, lucky, ok, perfect, positive, productive
2	Love	accepted, affectionate, beloved, bless, compassionate, delicate, devote, face, faithful, fond, hand, horny, hot, longing, lovely, loyal, naughty, nostalgic, old
3	Anger	agitate, annoyed, bitchy, bitter, bother, cold, cranky, dangerous, disgust, dissatisfied, distract, envious, frustrate, fuck, greedy, grouchy, grumpy, hat
4	Fear	afraid, agitate, anxious, apprehensive, confuse, distraught, distressed, doubtful, fear, fearful, frantic, frighten, helpless, hesitant, insecure
5	Surprise	body, completely, confuse, curious, daze, enthral, face, funny, head, impressed, learn, month, overwhelmed, see, sense, shock, sit, slightly, sort, story, strange

METHODOLOGY

3

1

Logistic regression

2

Method improvement

METHODOLOGY

Logistic regression

Tokenize input data

TF-IDF (t,d)= TF(t,d) * DF (t,d)

$$TF\ (t,d)\ = \frac{Number\ of\ times\ term\ t\ appear\ in\ document\ d}{Total\ of\ term\ in\ document\ d}$$

$$IDF\ (t)\ = log\ (\frac{total\ number\ of\ document}{number\ of\ document\ contain\ term\ d})$$

Stochastic gradient descent

$$w = w- \quad wLoss(xi,yi)$$

METHODOLOGY

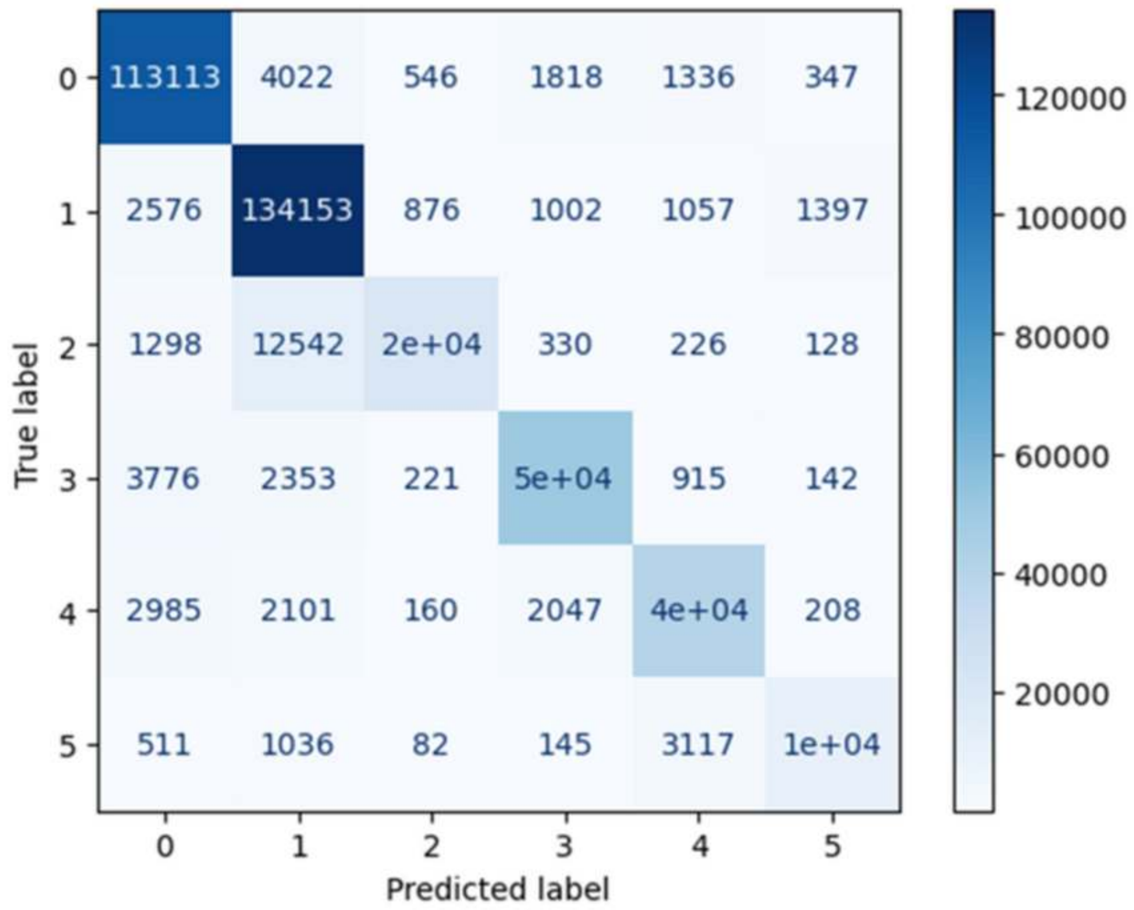
Logistic regression

```
Classification Report:
              precision    recall  f1-score   support

           0       0.91      0.93      0.92    121182
           1       0.86      0.95      0.90    141061
           2       0.91      0.58      0.71     34554
           3       0.90      0.87      0.89     57306
           4       0.86      0.84      0.85     47706
           5       0.82      0.67      0.74     14972

    accuracy                           0.88    416781
   macro avg       0.88      0.81      0.83    416781
weighted avg       0.88      0.88      0.88    416781

Accuracy:  0.8817
Precision: 0.8830
Recall:    0.8817
F1 Score:  0.8781
```



METHODOLOGY

Method improvement

Improve tokenization

$$TF(t,d) = 1 + log(TF_{raw}(t,d))$$

→ Validate the best hyperparameter → Modified model

EVALUATION

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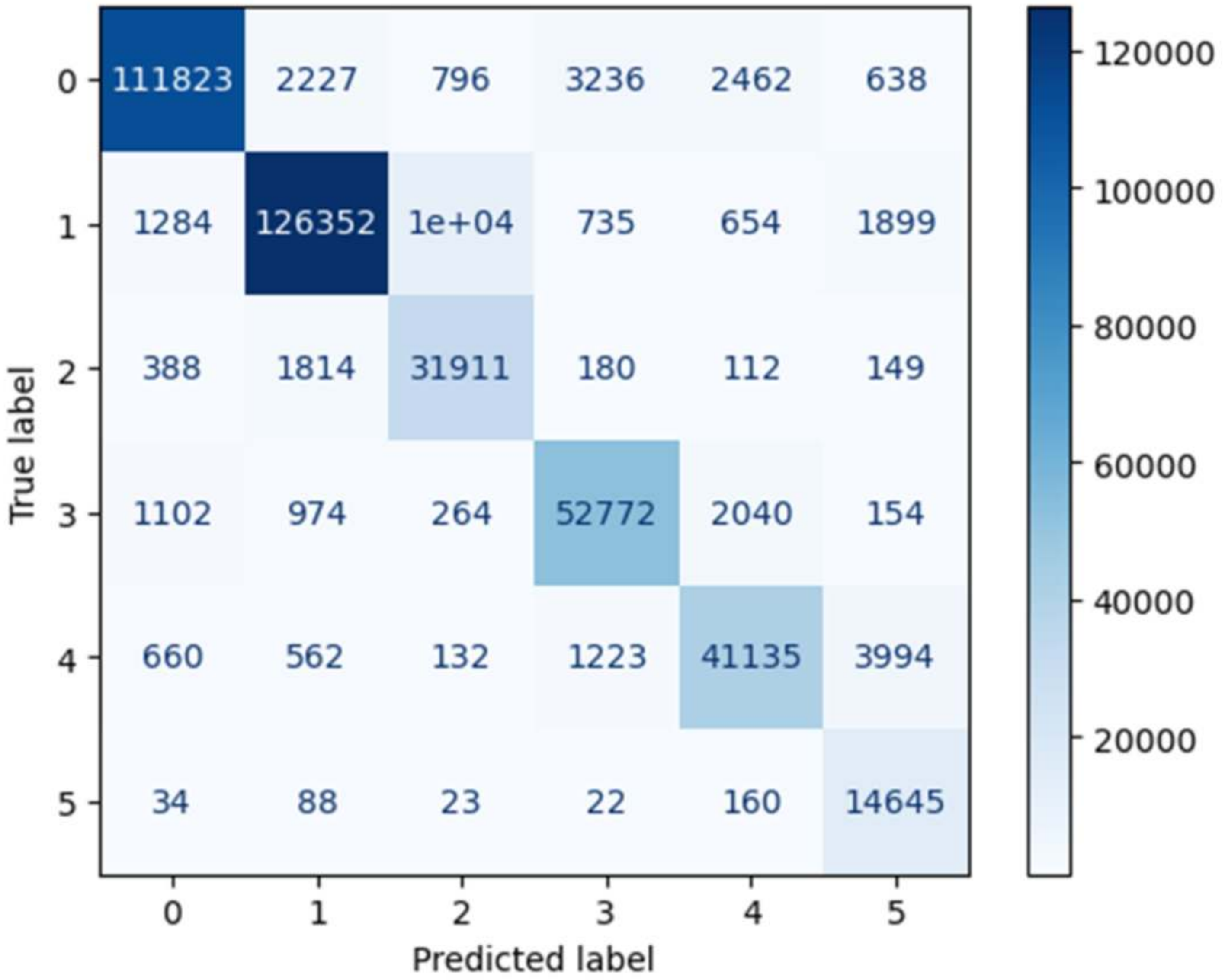
...
Classification Report:
              precision    recall  f1-score   support

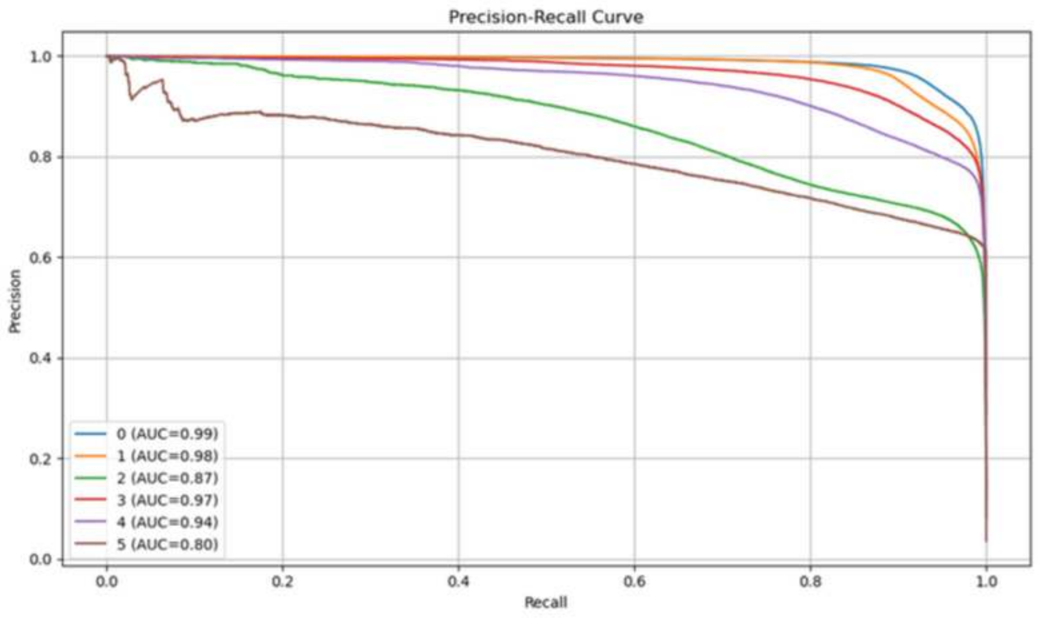
           0       0.97      0.92      0.95    121182
           1       0.96      0.90      0.93    141061
           2       0.74      0.92      0.82     34554
           3       0.91      0.92      0.91     57306
           4       0.88      0.86      0.87     47706
           5       0.68      0.98      0.80     14972

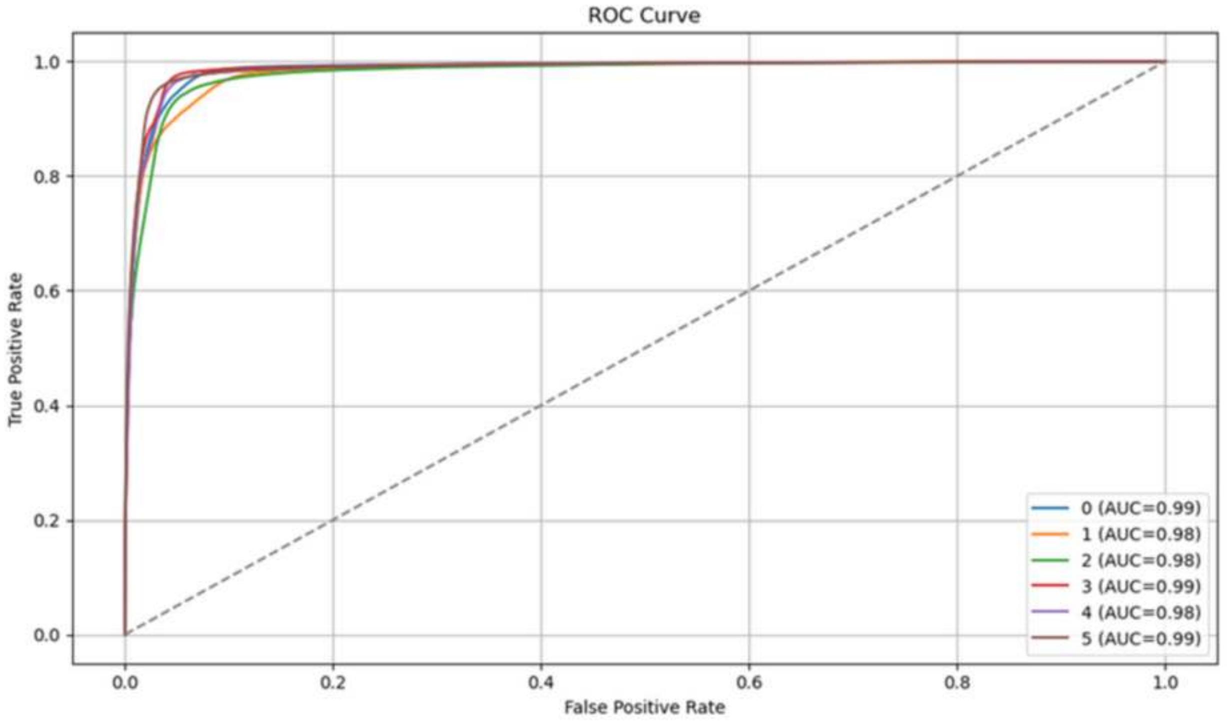
    accuracy                           0.91    416781
   macro avg       0.86      0.92      0.88    416781
weighted avg       0.92      0.91      0.91    416781

Accuracy:  0.9085
Precision: 0.9174
Recall:    0.9085
F1 Score:  0.9106
```



EVALUATION


Precision-Recall Curve
Precision
Recall
0 (AUC=0.99)
1 (AUC=0.98)
2 (AUC=0.87)
3 (AUC=0.97)
4 (AUC=0.94)
5 (AUC=0.80)


ROC Curve
True Positive Rate
False Positive Rate
0 (AUC=0.99)
1 (AUC=0.98)
2 (AUC=0.98)
3 (AUC=0.99)
4 (AUC=0.98)
5 (AUC=0.99)

EVALUATION

3

1

Training time complexity: O(e.n.d)

2

Testing time complexity: O(n.d)

EVALUATION

Methods comparation

ComplementNB / 8k features

	precision	recall	f1-score
0	0.93	0.93	0.93
1	0.92	0.91	0.92
2	0.79	0.82	0.81
3	0.89	0.89	0.89
4	0.86	0.85	0.86
5	0.78	0.76	0.77
accuracy			0.89
macro avg	0.86	0.86	0.86
weighted avg	0.89	0.89	0.89

Performance of complementNB of 8000 features

MultinomialNB / 8k features

	precision	recall	f1-score
0	0.91	0.92	0.91
1	0.90	0.91	0.90
2	0.76	0.76	0.76
3	0.89	0.85	0.87
4	0.84	0.84	0.84
5	0.76	0.69	0.73
accuracy			0.88
macro avg	0.84	0.83	0.84
weighted avg	0.88	0.88	0.88

Performance of multinomialNB of 8000 features

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.92	0.95	121182
1	0.96	0.89	0.92	141061
2	0.72	0.96	0.83	34554
3	0.91	0.93	0.92	57306
4	0.87	0.88	0.88	47706
5	0.68	0.95	0.79	14972
accuracy			0.91	416781
macro avg	0.85	0.92	0.88	416781
weighted avg	0.92	0.91	0.91	416781

Precision: 0.9205
Recall: 0.9094
F1 Score: 0.9119

Performance of Light Gradient Boosting Machine

THANK YOU FOR YOUR ATTENTION